

Can foundation models improve peptide–HLA stability prediction?

A leakage-controlled negative result, and what it costs to establish

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Abstract

We ask whether pretrained protein language models improve prediction of peptide–HLA class I kinetic stability. On a leakage-controlled split of 28,166 measured half-lives, the answer is **no**: a one-hot multilayer perceptron over the peptide and the NetMHCpan pseudosequence reaches $\rho_{\text{pool}} = 0.780$ on validation and 0.806 on a test split read exactly once, and **no foundation-model configuration we tested matches it**. The negative survives every obvious objection — encoder depth, an $80\times$ scale sweep, LoRA fine-tuning, cross-chain attention, label budget, and three further model families including a published peptide–MHC model. We go further than the negative and explain it: ESM-2’s notion of which residues matter is *orthogonal* to this endpoint (rank correlation -0.03 against the trained model’s ablation profile); predicted structures are *correct* yet their geometry does not track half-life (Mantel $r = -0.111$ at 92% power for $r = 0.2$); and two unrelated routes to modelling the peptide–groove interaction — pretraining on 96M protein–protein interactions, or a 561,793-parameter attention head — buy the *same* $+0.16$ and both stop below the baseline. We report prediction intervals whose marginal coverage holds (90.5% at a 90% target) and whose per-allele coverage does not (68.8% to 100%), the empirical ceiling the task permits (≈ 0.823), and the compute bill (7.5 GPU-hours, \$15.26), of which 88% was spent on configurations that land inside seed noise or below a free baseline.

1 Problem setup and notation

Each record pairs an HLA class I groove sequence m with a peptide p and an experimentally measured complex half-life $t_{1/2}$ in hours. Models predict the log-transformed target

$$y = \log(1 + t_{1/2}), \quad \hat{t}_{1/2} = \exp(\hat{y}) - 1,$$

which is used because the raw distribution spans 0 to 256.7 h with a median of 1.1 h and a point mass of 5,679 exact zeros (20.2% of rows).

Notation is fixed for the whole report.

Symbol	Domain	Meaning
p	Σ^9	peptide, 9-mer over the 20-letter amino-acid alphabet Σ
m	Σ^{182}	HLA class I $\alpha_1\alpha_2$ groove domain
$\pi(m)$	Σ^{34}	NetMHCpan pseudosequence: the 34 crystallographic contact positions
a	$\{1, \dots, 75\}$	allele identity, as a categorical label
$t_{1/2}$	$[0, 256.7]$	measured half-life, hours
y	$\mathbb{R}_{\geq 0}$	$\log(1 + t_{1/2})$, the regression target
$c(p)$	$\{1, \dots, 5410\}$	peptide cluster id under Hamming distance ≤ 2
E_θ	$\Sigma^L \rightarrow \mathbb{R}^{L \times d}$	frozen encoder producing per-residue states
d	$\mathbb{N}$	encoder width (320, 480, 640, 1280 across models tested)
ρ_{pool}	$[-1, 1]$	Spearman correlation over all rows pooled
ρ_{within}	$[-1, 1]$	mean Spearman inside each allele with ≥ 12 rows

Table 1: Notation. ρ_{within} is the primary metric; Section 3 explains why.

2 Data and the leakage-controlled split

2.1 Why a row split is wrong here

94% of peptides in this dataset are measured against more than one allele. A random split over *rows* therefore places the same peptide on both sides of the partition, and a model can score well by recognising the fragment rather than by modelling the interaction. Grouping by *exact* peptide removes that, but not the weaker form: a 9-mer differing by one substitution is 89% identical, and the field’s common 80%-identity convention admits exactly such pairs.

We therefore cluster peptides at **Hamming distance** ≤ 2 by single-linkage and assign whole clusters, which is strictly more conservative than both alternatives.

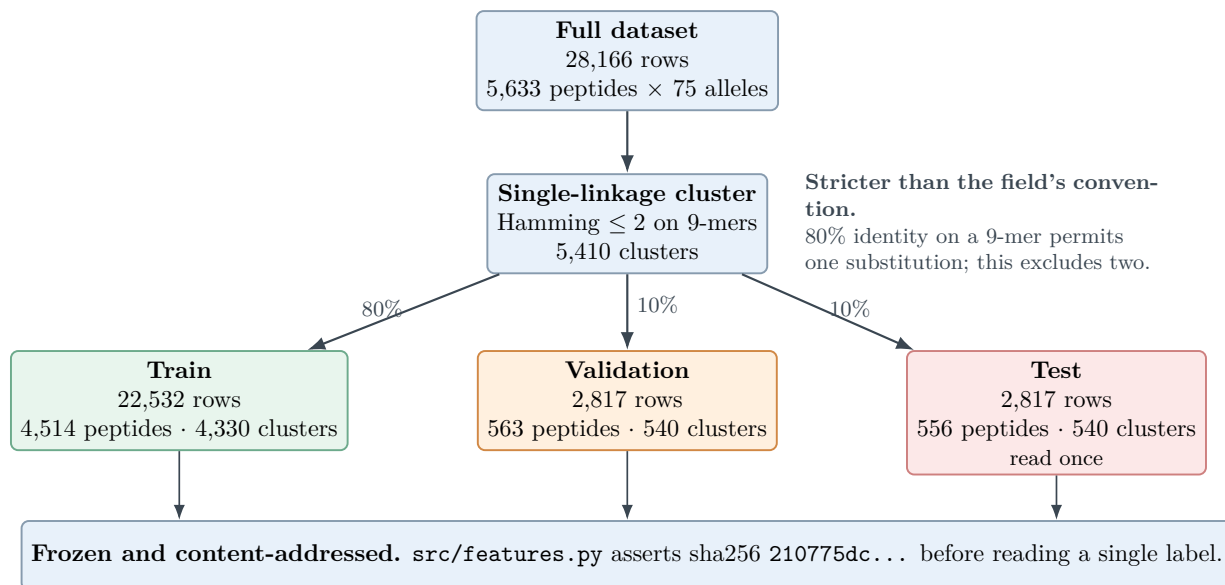


Figure 1: The leakage-controlled split. Whole clusters are assigned, so no peptide in one partition is within two substitutions of any peptide in another. The hash guard means a job that somehow receives a different split fails immediately rather than quietly reporting a number against other data.

2.2 The split was validated, not assumed

Two checks run against the frozen assignments. First, a **boundary probe**: a synthetic peptide 2 edits from a training peptide must be detected as a near-duplicate, and one at 3 edits, verified clear of *every* training peptide, must not be. Both behave correctly, which establishes the detector separates rather than merely firing. Second, measured **cross-allele agreement decays monotonically with pseudosequence distance** — 0.823, 0.659, 0.474, 0.318 at Hamming 1, 2, 3, 4 — with no model involved, confirming that sequence distance carries the biological signal the split assumes it does.

3 Evaluation protocol

3.1 A floor, not zero

A per-allele median containing **no peptide information whatsoever** scores $\rho_{\text{pool}} = 0.563$. Every pooled figure in this field must be read against that number rather than against zero, because pooled Spearman rewards a model merely for knowing which allele it is looking at. This is not a small correction: it is 72% of the way from zero to our best model.

3.2 Within-allele Spearman is primary

ρ_{within} averages the Spearman correlation computed *inside* each allele with at least 12 rows. It is the metric that cannot be inflated by across-allele structure, and it is also the clinical question — ranking

one patient’s candidate peptides within the allele they actually carry. Section 8 gives a case where the two metrics disagree so sharply that the choice decides whether a model appears useless or useful.

3.3 Intervals from a cluster bootstrap

All confidence intervals come from 2,000 paired bootstrap draws (seed 2026) resampling the **540 validation peptide clusters**, never rows. Rows inside one cluster are within two substitutions and are not independent; resampling them would claim precision we have not earned.

3.4 A negative control that found something

Shuffling training labels scored +0.104, which looks like leakage. It is not. A fixed-epoch rerun scored +0.02, isolating the remaining +0.08 as **epoch-selection optimism**: the act of choosing the best epoch against validation inflates every validation figure by roughly that much. Paired differences cancel it, which retroactively justifies reporting comparisons rather than absolute values.

3.5 Test discipline

The test split was read **once**, after 21 runs had frozen every architecture, budget and selection rule. A guard file in `src/test_evaluation.py` refuses a second read. One model (L150) was deliberately omitted from that read because scoring it would have required retraining *after* seeing the test table; it was closed later by keeping the validation-selected checkpoint and scoring inside the same run (Section 6).

4 Models

4.1 The ladder

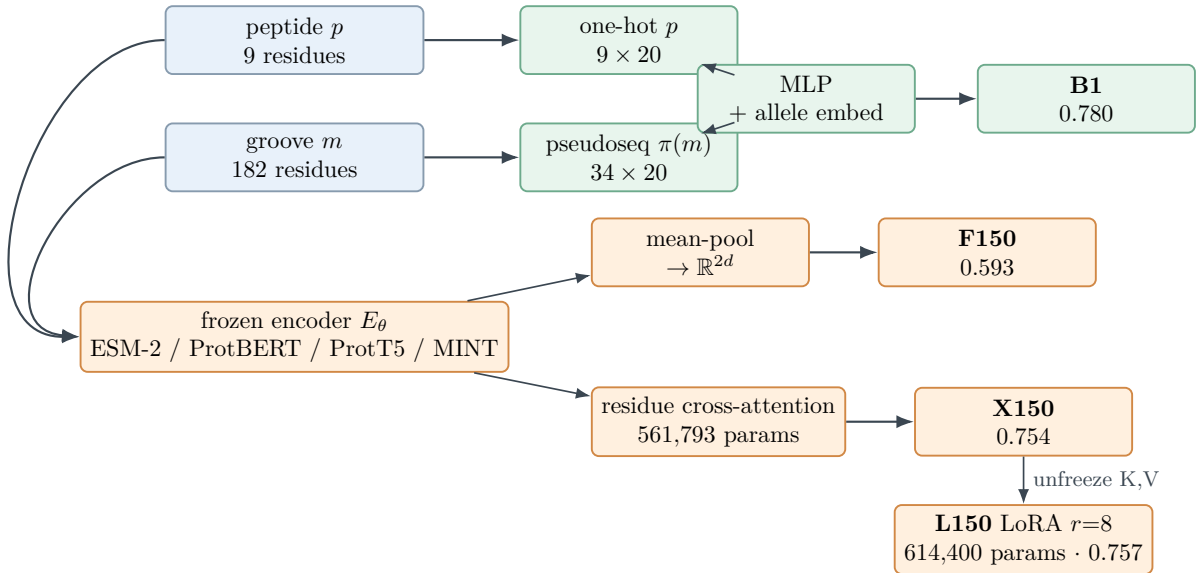


Figure 2: The two paths. Everything below the encoder is a protein language model; the top path has none and wins. Figures are validation ρ_{pool} .

4.2 What each rung isolates

Rung	Trainable	The single question it answers
B0b	—	How much of pooled ρ is available with <i>no</i> peptide information?
B1'	MLP	Does a one-hot allele <i>label</i> suffice, without its sequence?
B1	MLP	Does the crystallography-derived pseudosequence add over the label?
F150	head	What does a frozen PLM give when position is destroyed by pooling?
X150	561,793	What does the <i>head</i> contribute on identical frozen embeddings?
L150	614,400	Does adapting the encoder itself (LoRA on K,V of all 30 layers) help?

Table 2: Each rung changes exactly one thing from the rung above it.

LoRA uses the standard reparameterisation [2] $W_{\text{eff}} = W + \frac{\alpha}{r}BA$ with $r = 8$, $\alpha = 16$, dropout 0.05, applied to the key and value projections of all 30 layers: 60 modules, 614,400 parameters, 0.41% of the backbone.

4.3 Fair comparison across encoders

Every encoder — ESM-2 at four sizes [5], ProtBERT and ProtT5 [1], and MINT [12] — is run through the **identical** head, split, three seeds, early-stopping budget (40 epochs, patience 6), and a two-learning-rate choice (10^{-3} , 3×10^{-4}) with the better kept. Only the encoder changes. This matters: an early version of our scale sweep gave ESM-2 650M a single learning rate and recorded 0.526, *below the no-peptide floor*. That was a convergence failure, not a finding; the fair two-rate protocol gives 0.764.

5 Compute setup and dataflow

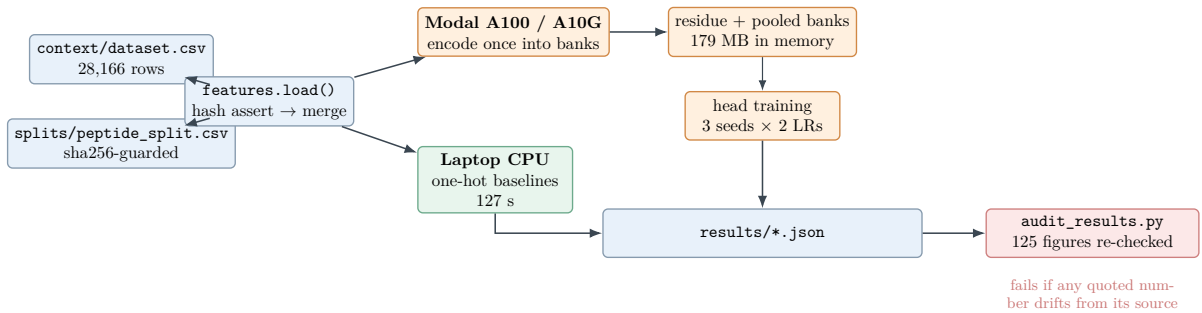


Figure 3: Dataflow. Encoding happens once into shared banks indexed per row; materialising per-row tensors instead would have required 10.5 GB, where the banks occupy a measured 179 MB.

6 Results

6.1 The ladder

Rung	ρ_{pool} [95% CI]	ρ_{within} [95% CI]	PLM?
B0b per-allele median (no peptide info)	0.563 [0.522, 0.598]	undefined	no
F150 frozen ESM-2 150M, mean-pooled	0.593 [0.556, 0.625]	0.278 [0.222, 0.320]	yes
ProtT5-XL encoder, 1,208M	0.729 [†]	0.511 [†]	yes
ProtBERT, 420M	0.729 [†]	0.529 [†]	yes
MINT Stage-1, 814M, mean-pooled	0.749 [†]	0.528 [†]	yes
B1' one-hot peptide + allele identity	0.748 [0.719, 0.775]	0.557 [0.508, 0.585]	no
X150 frozen ESM-2 150M, cross-attention	0.754 [0.727, 0.777]	0.558 [0.508, 0.585]	yes
L150 ESM-2 150M, LoRA-adapted	0.757 [0.732, 0.780]	0.572 [0.524, 0.599]	yes
ESM-2 650M, fair two-LR protocol	0.764 [†]	—	yes
B1 one-hot + pseudosequence + allele	0.780 [0.756, 0.802]	0.633 [0.588, 0.654]	no

Table 3: Validation. [†] not bootstrapped; worst per-seed spread across those rows is 0.039. B1 – L150 = +0.023 [+0.007, +0.039] pooled and +0.060 [+0.027, +0.094] within-allele; both exclude zero.

6.2 Held-out test, read once

Rung	test ρ_{pool}	test ρ_{within}
B0b per-allele median	0.573	undefined
F150 frozen ESM-2, mean-pooled	0.606	0.244
X150 frozen ESM-2, cross-attention	0.756	0.535
L150 LoRA-adapted ESM-2	0.758	0.525
B1' one-hot + allele identity	0.767	0.566
B1 one-hot + pseudosequence + allele	0.806	0.645

Table 4: Every ordering survives and the gap *widens*. B1 – X150 is +0.050 pooled on test against +0.026 on validation; B1 – L150 is +0.048 and +0.120, roughly double its validation value.

6.3 Every obvious objection, tested

Objection	Result	Against
“Use a deeper layer” — five ESM-2 layers	0.108 range	worst seed spread 0.157: noise
“Use a bigger model” — 8M to 650M, 80×	0.007 range	worst seed spread 0.036: noise
“Fine-tune it” — LoRA on K,V	+0.003 pooled	CI [−0.012, +0.018]: spans zero
“It needs fewer labels” — 5% to 100%	no crossover at any budget	gap <i>widest</i> at 25%
“Let it see both chains” — joint 191-mer	−0.017 pooled	within noise
“Use a different family” — ProtBERT, ProtT5	both 0.729	below ESM-2 <i>and</i> below B1
“Use a published pMHC model” — MINT	0.749	−0.031 against B1
“Ensemble it”	+0.009 [+0.001, +0.017]	real, and tiny

Table 5: The data-efficiency result runs *backwards*: scarcity makes the foundation model more disadvantaged, not less.

7 Why it fails: four independent measurements

7.1 ESM-2’s salience is orthogonal to the endpoint

Ablating each peptide position in the trained model yields a sharply peaked profile — $P9 \gg P2 > P1 > P3$, with $P4$ – $P8$ contributing almost nothing. That is the textbook B-pocket/F-pocket anchor picture [9], recovered from data alone and stable across seeds. ESM-2’s own masked-position likelihood

profile over the same peptides is **flat**, and the two rank-correlate at -0.03 . Its per-peptide pseudo-log-likelihood predicts stability at $\rho_{\text{pool}} = +0.002$.

7.2 The groove does not rescue the likelihood

Scoring each peptide *inside* the 191-residue peptide–HLA concatenation rather than alone moves the correlation from 0.002 to 0.001; the difference between the two scores predicts nothing (0.004). The informative detail is that the two scores rank-correlate at only **0.669**: the groove *does* substantially re-order which peptides ESM-2 finds likely. Something real happens, and it is uncorrelated with stability. We state the limit rather than glossing it: ESM-2 has no chain-break token, so this cannot separate *the groove explains nothing* from *ESM-2 cannot condition on a chain it does not know is separate*.

7.3 The interaction must be modelled by something trained to model it

This was the weakest claim in our earlier write-ups, resting on a comparison across two studies with different splits. It is now a measurement on ours.

Route to the interaction	Mechanism	$\Delta\rho_{\text{pool}}$	$\Delta\rho_{\text{within}}$
Cross-attention head on frozen ESM-2	561,793 trainable parameters	+0.161	+0.276
MINT cross-chain pretraining	96M protein–protein interactions	+0.156	+0.250
ESM-2 merely <i>permitted</i> to attend across chains	none	−0.017	within noise

Table 6: Both deltas are measured against F150, the architecturally matched mean-pooled comparator. Two unrelated routes agree to within 0.005; self-attention handed the opportunity without interaction training discovers nothing.

MINT mean-pooled reaches 0.749 and X150 reaches 0.754 — a difference of 0.005, inside seed spread. **Two independent mechanisms converge on the same ceiling, and one-hot sits above both at 0.780.**

7.4 Correct structures, no usable signal

Boltz-2 [7] poses *do* depend on the peptide: across-peptide variation is $4.1\times$ the model’s own diffusion noise, so the pessimistic hypothesis — one canonical pose regardless of peptide — is false. The poses are also right: Boltz-2 recovers **31 of the 31** pseudosequence positions present in our domain from predicted structure alone, against 16.5 expected by chance (label-permutation $p = 0.0001$), with none missed. Yet that geometric variation does *not* track half-life: Mantel $r = -0.111$, $p = 0.116$, permuting peptide labels, with **92% power to detect a true r of 0.2** [6]. The structural arm is a negative with the power to mean it.

8 The metric choice is load-bearing

MINT Stage-1 [12, 3] is a released 814M model built for peptide–MHC class I: the MINT backbone fine-tuned on 126K NetMHCpan 4.1 binding affinities [11]. Crucially **it has never seen a half-life**. Scored zero-shot, with no training of any kind:

	ρ_{pool}	ρ_{within}
MINT Stage-1, zero-shot	0.523	0.421
B0b per-allele median, <i>no peptide information</i>	0.563	undefined
B1 one-hot, trained	0.780	0.633

Table 7: Read pooled alone and the model appears to have no signal, falling below the no-peptide floor. Read within-allele and it has a great deal, from a model that never saw this endpoint.

Both numbers are correct. A zero-shot affinity score has no reason to be calibrated *across* alleles against half-life, because alleles differ systematically in their stability distributions and an IC_{50} -trained scalar does not encode that. The across-allele component is close to noise and drags the pooled figure beneath a floor that receives allele identity for free; the within-allele ranking survives intact. This is the clearest justification in our work for treating ρ_{within} as primary — here the metric choice is the whole difference between “*this model does nothing*” and “*this model carries real signal with the wrong calibration*”. It is also precisely the gap that SPEARMINT’s assay-conditioned FiLM head [8] exists to close.

An objection pre-registered, not answered afterwards. MINT documents its input as a full ≈ 365 -residue heavy chain; our dataset carries the 182-residue groove. A truncation penalty would masquerade as “MINT does not work here”, so both conditions were run before either was examined: 182 residues, and the same sequence extended by a constant α_3 domain, which is near-invariant across class I alleles and so can only change length, never allele-specific signal. **The difference is 0.0012.** Length is not the explanation.

9 Which comparators are admissible

Model availability and model admissibility are different questions, and conflating them would have produced a flattering but meaningless comparison.

Model	Fine-tuned on	Admissible here
NetMHCstabpan [10]	this dataset	no — leakage
SPEARMINT (Stage 3) [3]	pMHC stability	no — same endpoint
mint-stage2-stability	pMHC stability	no — same endpoint
mint-stage1-affinity	pMHC affinity only	yes

Table 8: All four are downloadable today. Three are excluded because a model trained on the endpoint we are predicting measures memorisation rather than skill, and we could not prove otherwise.

10 Uncertainty: intervals that state their own limits

Split-conformal calibration [4] uses 3,412 rows carved from train as **whole peptide clusters**. This detail is what makes the guarantee valid: calibration rows drawn at random would share clusters with rows the model trained on, so their residuals would measure *already-seen* difficulty while validation and test rows all come from unseen clusters. The intervals would come out systematically too narrow and the coverage guarantee would fail *silently*, because nothing errors and coverage merely falls short.

Target	q (log scale)	val coverage	test coverage	median width (h)
80%	0.773	79.2%	80.2%	4.22
90%	1.103	89.9%	90.5%	6.63
95%	1.422	95.3%	95.4%	9.65

Table 9: Marginal coverage holds to within half a point at every level, using the finite-sample quantile $\lceil (n+1)(1-\alpha) \rceil / n$.

Conditional coverage does not hold, and reporting only the marginal number would have been the dishonest version. At a 90% marginal target, per-allele coverage across 68 test alleles has median 92.6% and ranges from **68.8% to 100%**, with 7 of 68 alleles below 80%. An interval advertised as 90% gives HLA-A*01:01 patients 68.8%. Conformal guarantees coverage averaged over rows and nothing within a subgroup; restoring per-allele guarantees requires Mondrian conformal and more calibration rows than the smallest alleles possess.

11 What it cost

Approach	$\rho_{\text{pool}}/\rho_{\text{within}}$	Compute	Cost
B1 one-hot	0.780 / 0.633	127 CPU-seconds, laptop	\$0.00
Best single ESM-2	0.764 — still below	A100	\$6.70
L150 LoRA (trained twice)	0.757 / 0.572	22,786 A100-seconds	\$13.28
One-hot + ESM-2 ensemble	0.789 / 0.639	A10G	\$0.11 for +0.009
Whole project	—	7.5 GPU-hours	\$15.26

Table 10: 88% of the GPU bill went to configurations that land inside seed noise or below a free baseline. The model that wins costs nothing and runs on a laptop.

Where the ESM models *do* win is worth stating: on **median** absolute error (L150 0.87 h against B1 0.99 h) and on unstable complexes below 2 h (0.78 against 0.95 h). B1 leads on the 141 stable rows above 24 h, where errors are large enough to dominate the mean. A rank metric cannot see which regime the errors live in.

12 How much signal remains

Replicate agreement between measurements of the same peptide against different alleles gives an empirical ceiling of ≈ 0.823 . Our best model reaches 0.780 on validation and 0.806 on test, so roughly **0.19 of Spearman remains available** between the baseline and what the assay itself supports. The negative result is therefore not “this task is solved” but “this particular lever does not move it”.

13 Threats to validity

1. **Four families, not all of them.** ESM-2, ProtBERT, ProtT5 and MINT were tested. ESM-C has no official loadable checkpoint; SaProt requires foldseek 3Di structural tokens we could not build for 5,633 peptides. MINT’s *per-residue* states behind a residue cross-attention head remain untried and are the most likely configuration to exceed 0.75.
2. **The +0.156 in Table 6 is confounded.** MINT Stage-1 is both interaction-pretrained and affinity-supervised; separating the two requires base MINT, which is not published.
3. **Peptide overlap with NetMHCpan 4.1 is unverified.** MINT’s affinity training set likely shares peptides with our split. The labels it saw were affinities, never half-lives, so this is transfer learning rather than label leakage — but a familiarity advantage cannot be excluded without their training set.
4. **No hyper-parameter search.** Every model received the same budget and selection rule, which makes the comparison fair but leaves no model at its ceiling.
5. **Affinity pre-training on our own data was not attempted.** It is the largest published lever here and requires decontaminating the NetMHCpan corpus against our test clusters.
6. **One assay, 9-mers, 75 alleles.** Class I binds 8–11-mers; this is the 9-mer slice, and assay-to-assay shifts of the kind SPEARMINT recalibrates are not modelled.
7. **Validation figures carry epoch-selection optimism** of roughly +0.08; paired differences do not.
8. **We cannot compare against NetMHCstabpan**, because this dataset is its training data.

14 Reproducibility

- **Split:** `splits/peptide_split.csv`, sha256 210775dc4ad179df635b3386bdb9671e4bca515cef1892bc70fb38f9b1b73f47, asserted at load.
- **Runs:** `results/README.md` logs 30 runs, each with its exact command, what it establishes, and an explicit “*what this does not establish*” section.
- **Audit:** `scripts/audit_results.py` re-checks 125 quoted figures against the JSON that produced them, 11 prose restatements across five documents, and the front page’s own claims about how many runs and figures stand behind it. The last check exists because those counts had silently drifted to “21 runs” and “81 figures”.
- **Environment:** `requirements.txt` pins the local environment. This matters more than it sounds — B1 differs by 0.010 between Anaconda and PyPI scikit-learn, the same magnitude as several findings reported here. A known gap is stated rather than hidden: the Modal images still build unpinned.
- **External checkpoints** are pinned by commit, not by branch. MINT is loaded at 8bf8e51906cf63336706d6cfc4f85e95a8109c86; it ships custom modelling code, so an unpinned load would both execute whatever `main` holds and hold the model to a looser standard than the hash-guarded data.
- **Seeds:** 42, 43, 44 throughout; bootstrap seed 2026.

15 Conclusion

On this dataset, with leakage controlled at Hamming distance ≤ 2 , **a one-hot multilayer perceptron over the peptide and the NetMHCpan pseudosequence beats every protein language model configuration we tested**, and the margin widens on held-out data. The useful claim is not that foundation models do not work. It is narrower and more actionable: *a frozen single-chain protein language model — ESM-2, ProtBERT, ProtT5 or MINT alike — contributes about one hundredth of a Spearman point here, for several dollars and several GPU-hours, and we can show where the remaining signal is instead.*

What a foundation model *does* buy is the interaction, and only when it was trained to model one: cross-chain pretraining and a half-million-parameter attention head purchase the same +0.16, and both stop below a baseline that costs nothing. The 34 positions of the NetMHCpan pseudosequence — decades of crystallography compressed into a feature-selection decision — turn out to be a stronger prior than 96 million protein–protein interactions, and that is the result worth carrying forward.

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